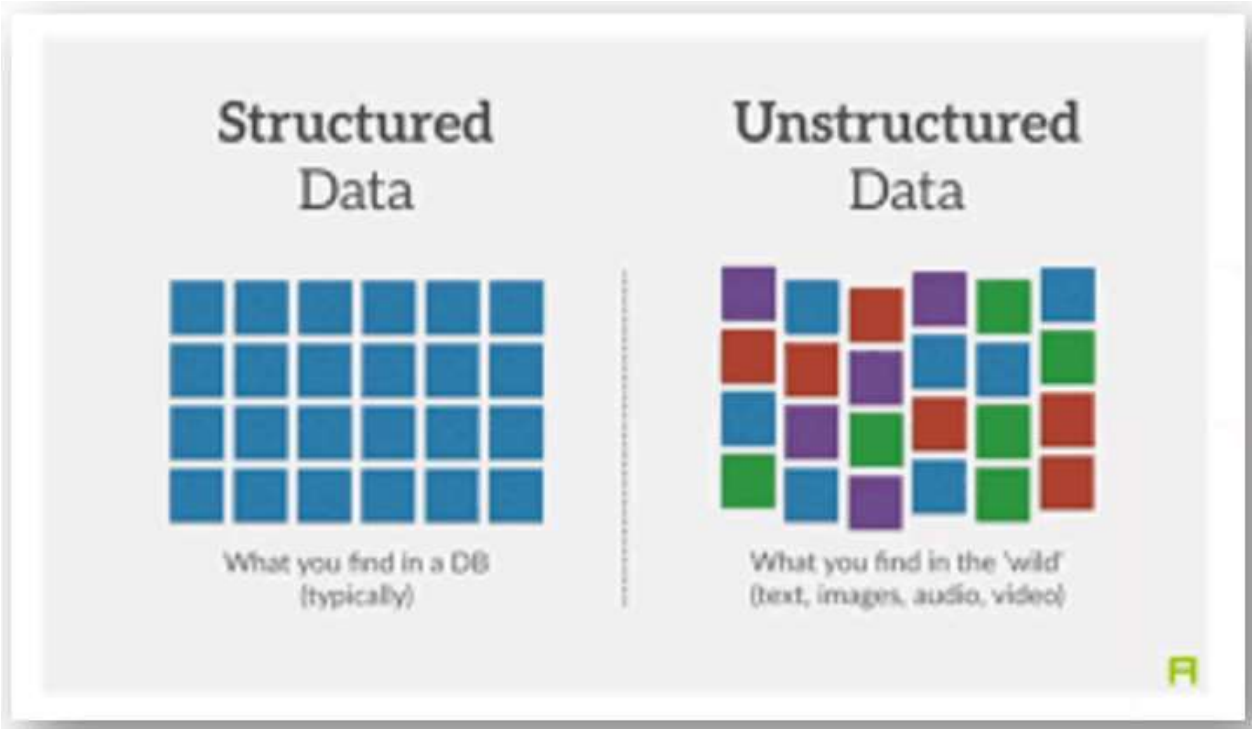
What is Data Science (DS)?

Data Science involves use of statistics, mathematics, computer science, and domain-specific fields to analyze and interpret complex data sets.



Working with Arrays with Numpy

• Numpy is a python library used to work with multi-dimention array

NumPy is the fundamental package for *scientific computing* in Python. It is a Python library that provides a multidimensional array object, various derived objects (such as masked arrays and matrices), and an assortment of routines for *fast operations on arrays*, including mathematical, logical, shape manipulation, sorting, etc.

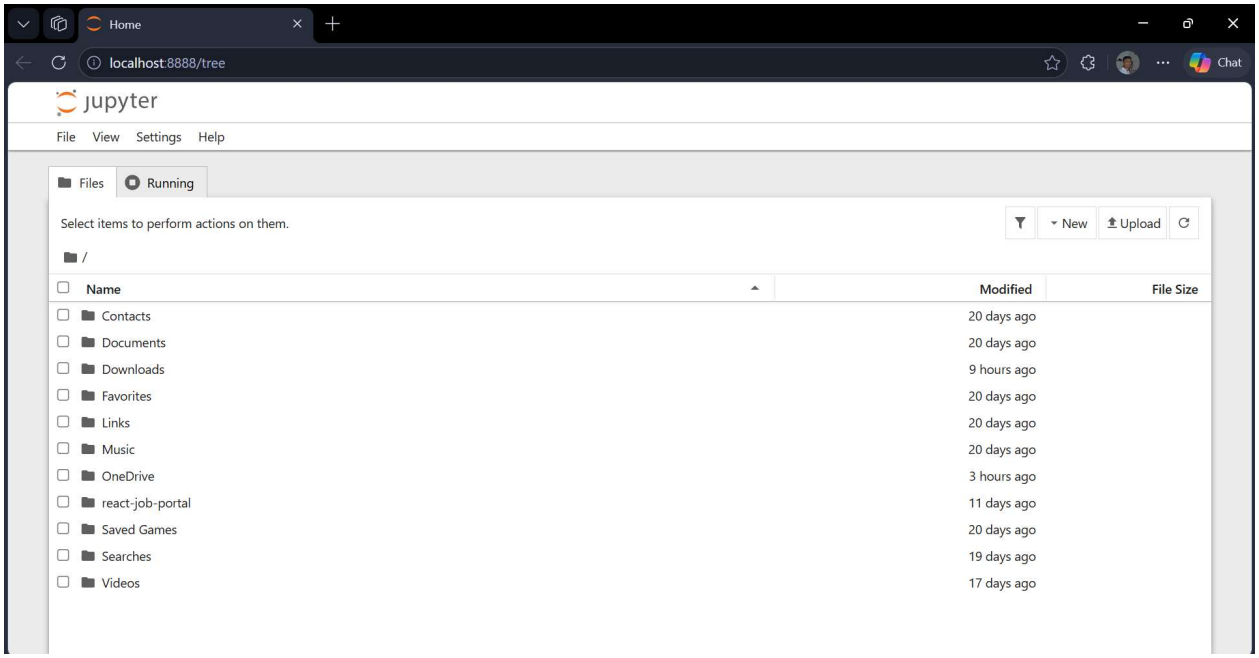
Install jupyter notebook with the command

Pip install notebook

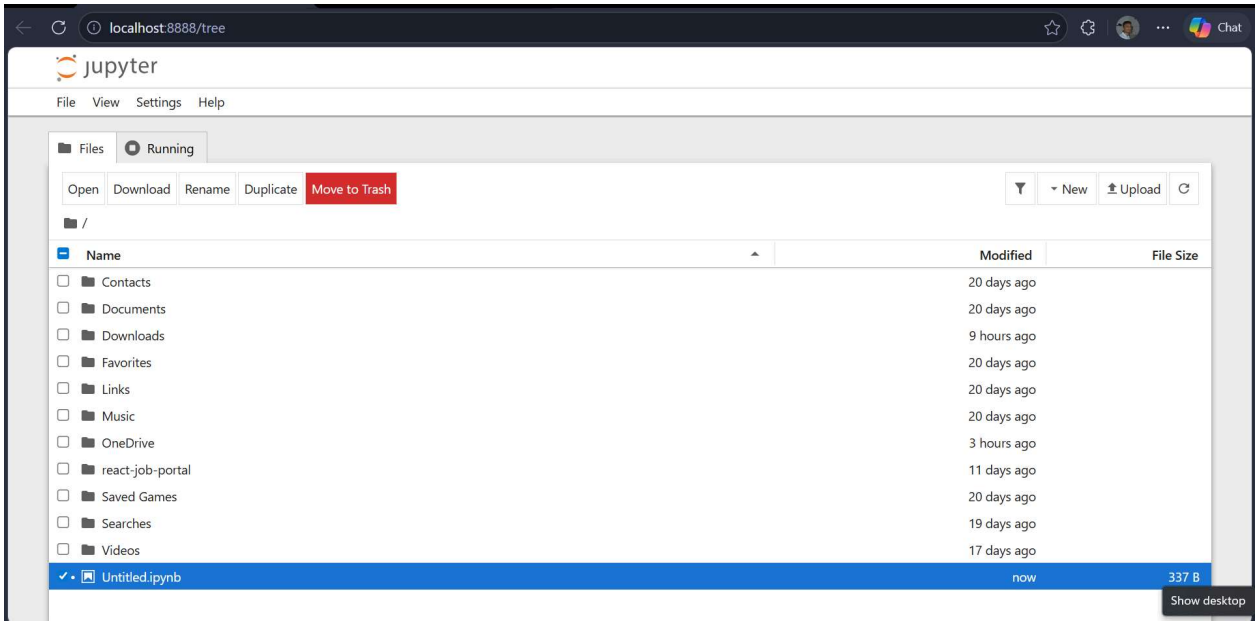
Then type

Python -m notebook

Your jupyter notebook page will open.

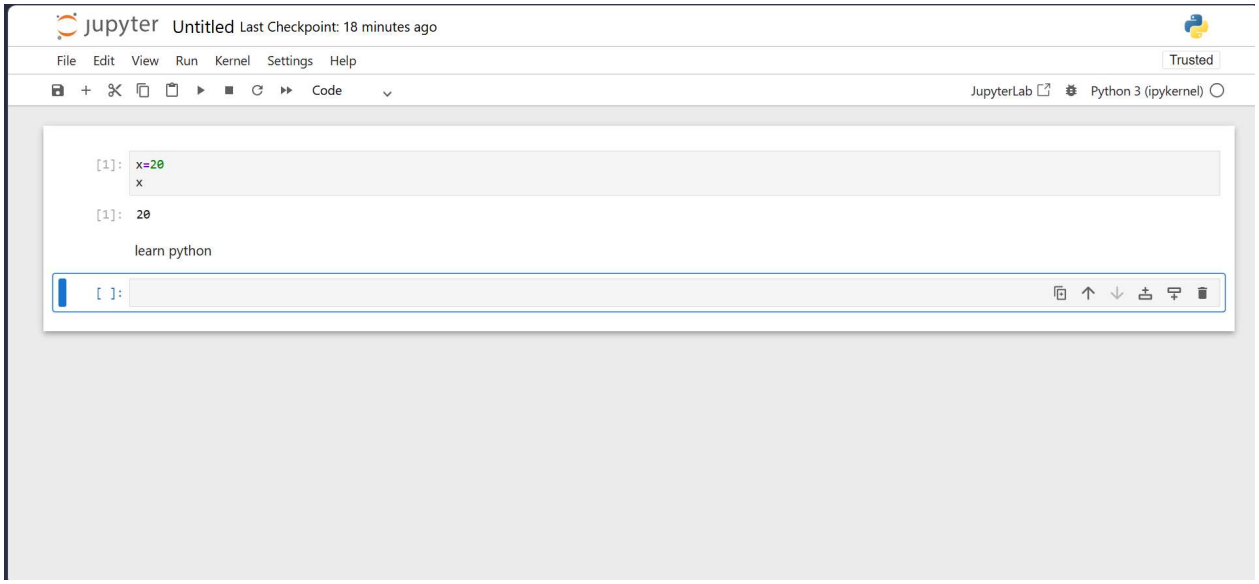


Click new , then python3 kernel,you will get your working page.

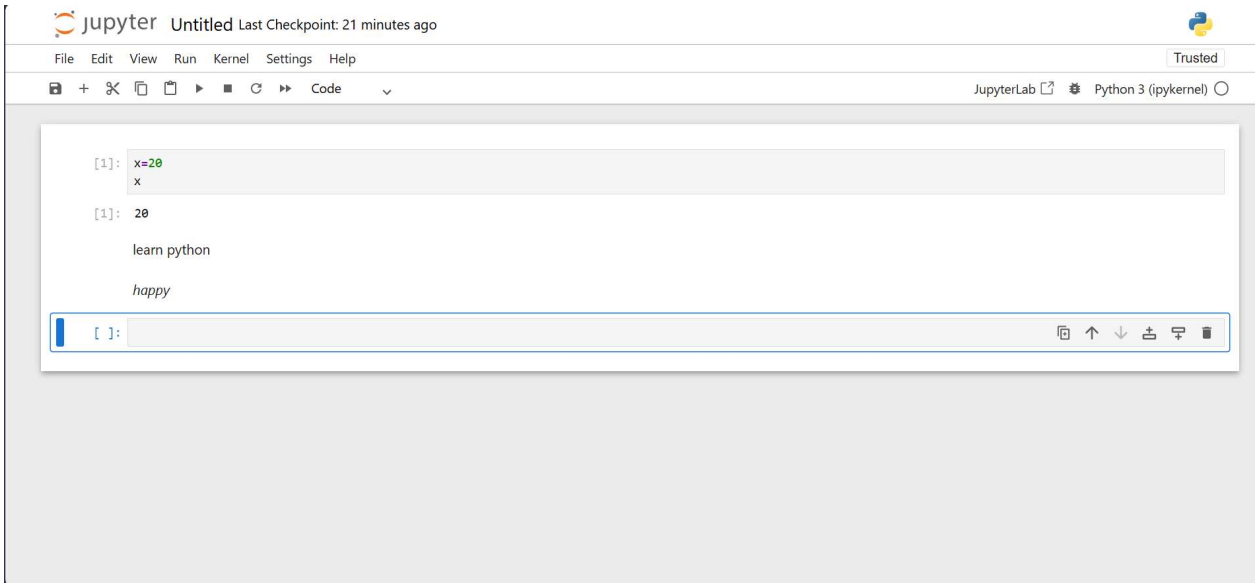


To add a heading,we can use markdown:-

Here we can use # sign to increase the size,like
->little big
->little more big
->A bit more
And so on



Any text inside *happy* ,for italics.



logout then, you need to kill the process in your terminal or command window

shutdown….it is all over.

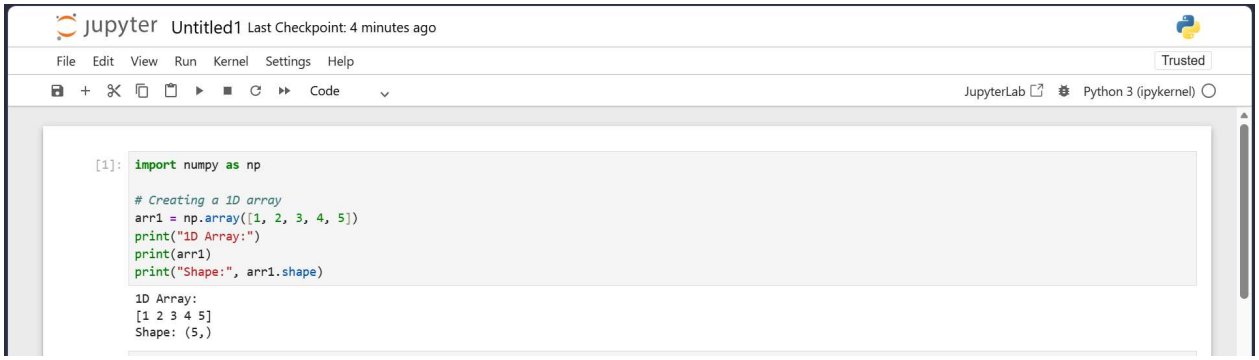
Note: When you start the notebook first time it will ask you to select the python kernel.

Install the numpy library.

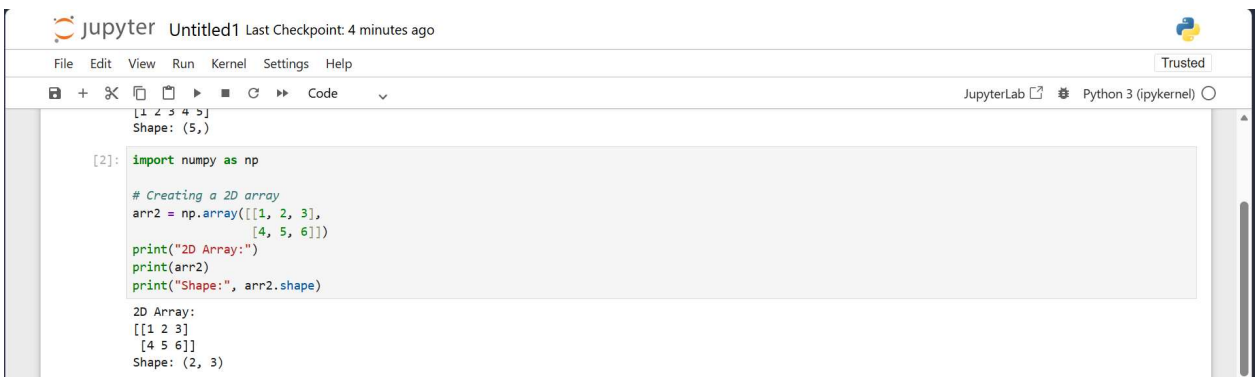
Open command prompt and type

pip install numpy

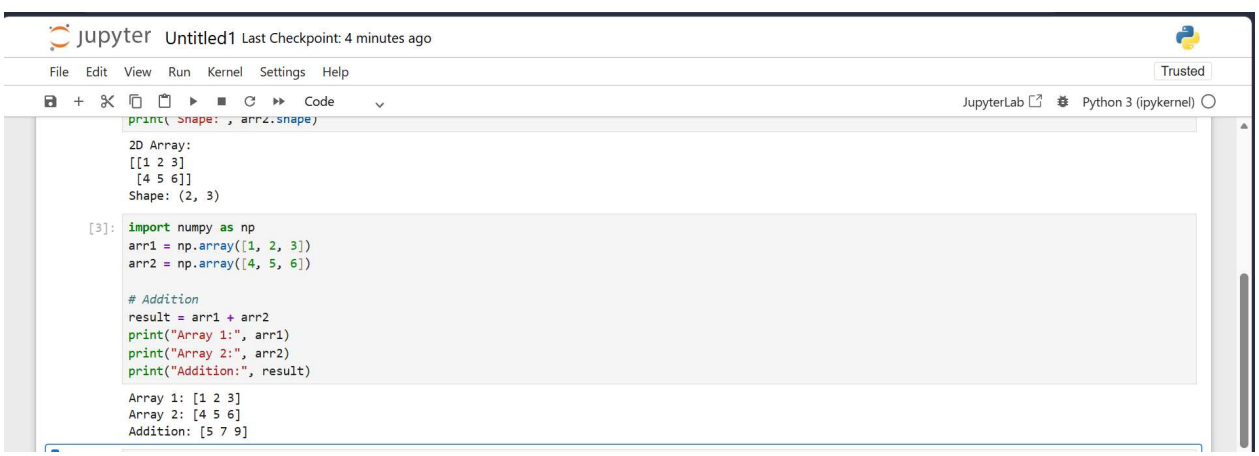
1-d array



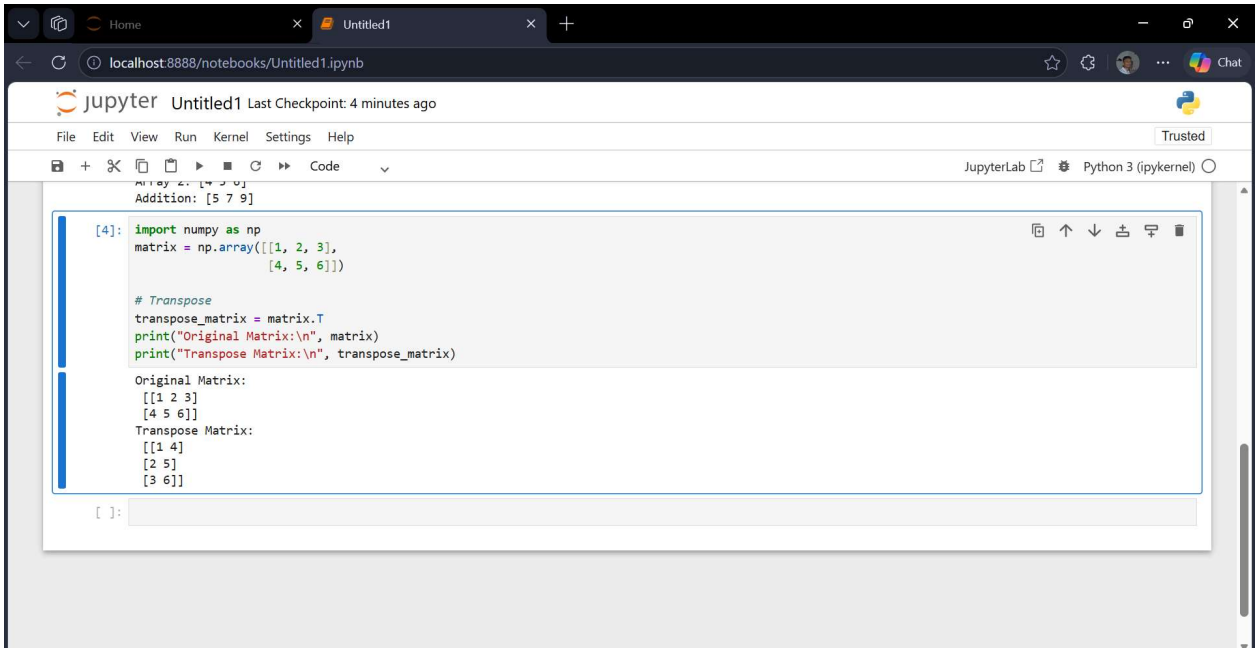
2-d array



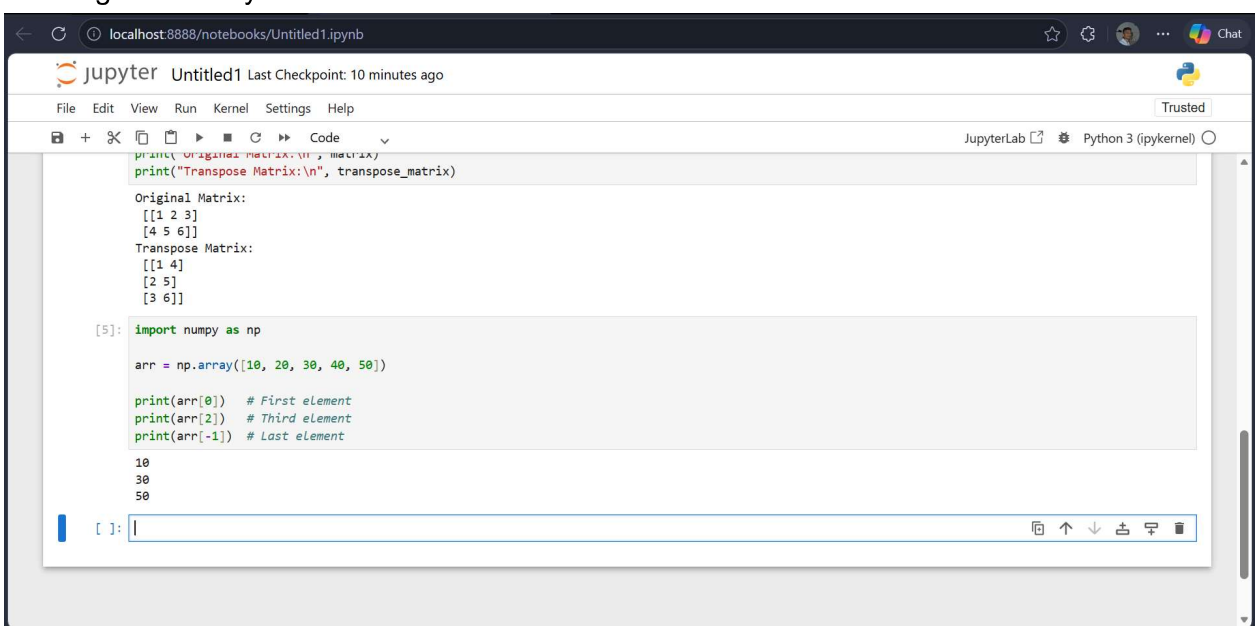
Addition of matrix:-



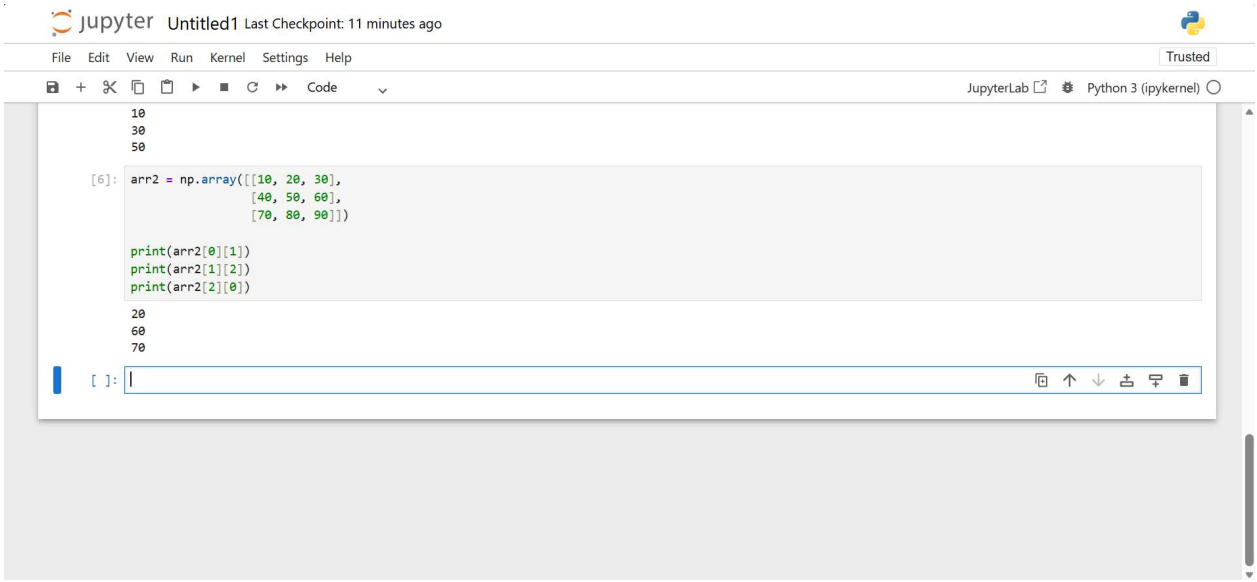
Transpose of a matrix:-



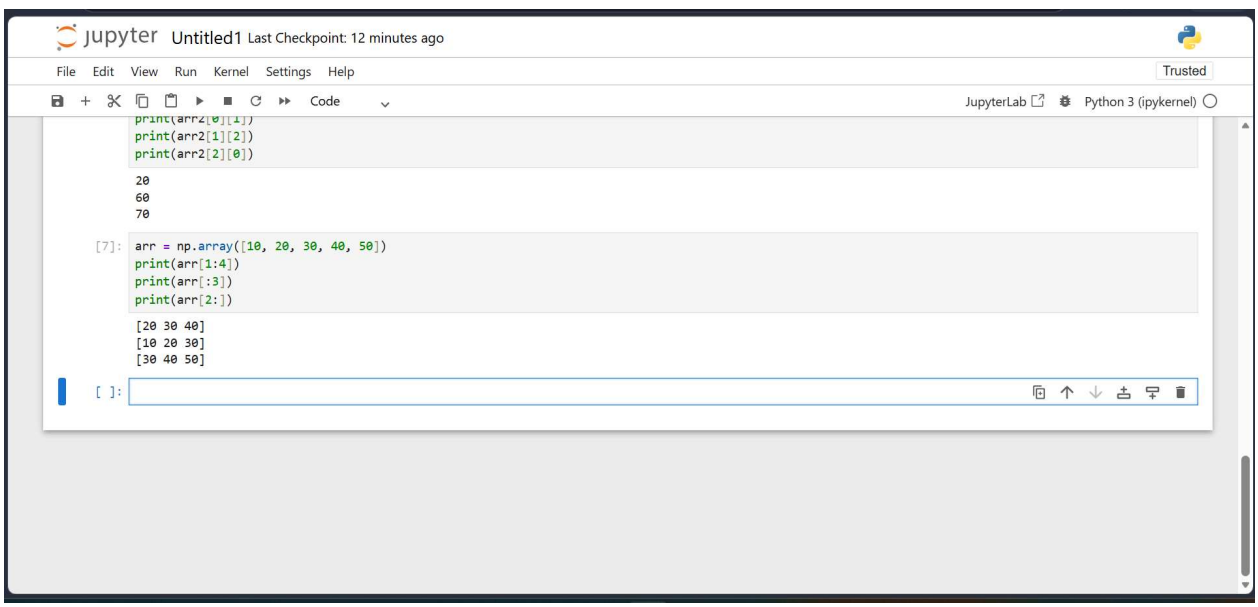
Indexing in 1d array:-



Indexing in 2d array:-



Slicing in 1d array:-



Slicing in 2d array:-

```
[20 30 40]
[10 20 30]
[30 40 50]
```

```
[8]: arr2 = np.array([[10, 20, 30],
                      [40, 50, 60],
                      [70, 80, 90]])

     print(arr2[0:2, 0:2])

     print(arr2[:, 1])
```

```
[[10 20]
 [40 50]]
[20 50 80]
```

Extracting values based on certain conditions:-

```
print(arr2[:, 1])
```

```
[[10 20]
 [40 50]]
[20 50 80]
```

```
[9]: myarray = np.array([[10,20,30],[40,50,60],[70,80,90]])

     user_condition = int(input('Enter the threshold value'))

     required_data = myarray > user_condition
     result_data = myarray[required_data]

     print('Data based on the condition that each value should be > 50')
     result_data
```

```
Enter the threshold value 25
Data based on the condition that each value should be > 50
[9]: array([30, 40, 50, 60, 70, 80, 90])
```